

Basic Network Hardware Theory

1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page. Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
```

Fig 1. Testing that both devices are communicating

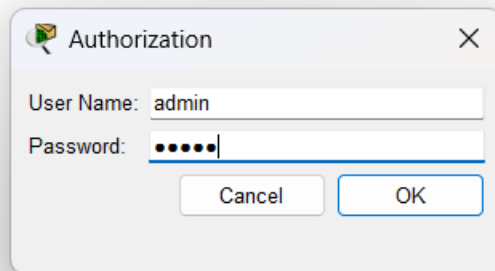


Fig 2. Authorization to access admin page of router

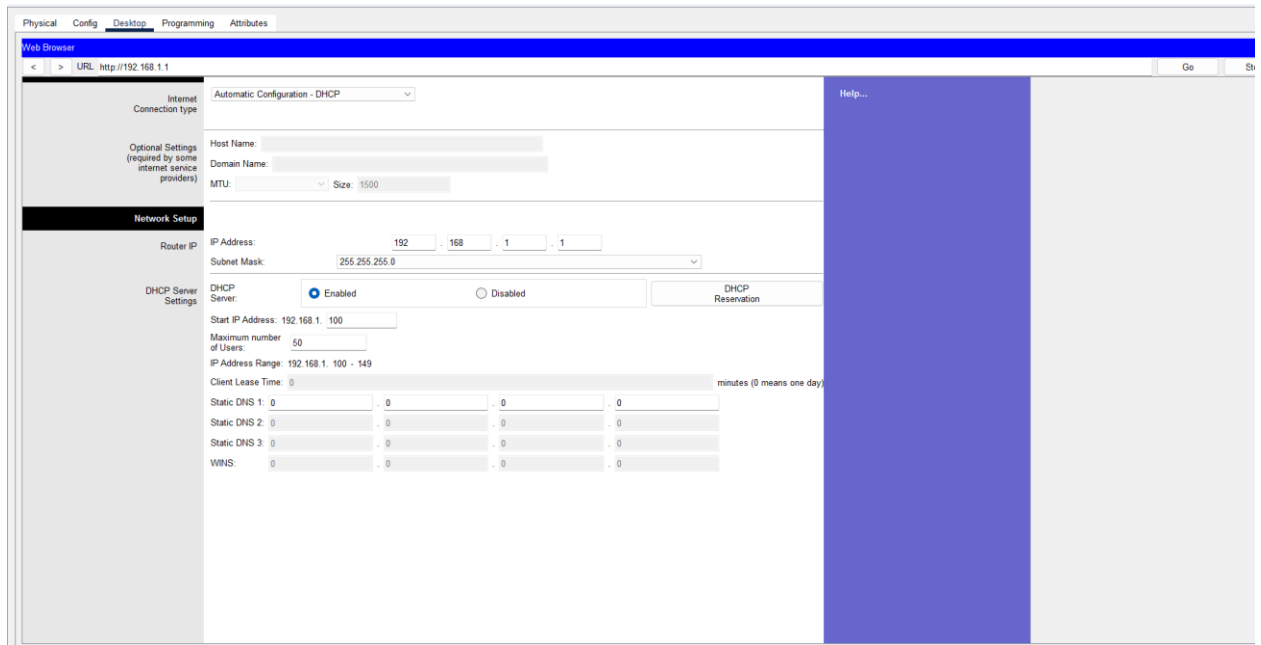


Fig 3. Router admin page

1. Connect Devices

- Connect **PC → Router** using Ethernet cable.

2. Set Router IP

- Router IP: 192.168.1.1
- Subnet Mask: 255.255.255.0

3. Configure PC IP

- Give PC a static IP:
- IP: 192.168.1.10
- Mask: 255.255.255.0
- Gateway: 192.168.1.1

4. Test Connection

- Open Command Prompt.
- Type: ping 192.168.1.1
- If you get "Reply", PC is connected to the router.

5. Open Router Page

- Open Web Browser.
- Enter: http://192.168.1.1

6. Login

- Enter router username and password.
- Open router settings page.

7. Check Settings

- Verify:
 - Router IP
 - DHCP status
 - Wireless name (SSID)

2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix  . :  
IPv6 Address. . . . . : 2409:40c1:5e:72a7:7c4f:ed70:184e:9475  
Temporary IPv6 Address. . . . . : 2409:40c1:5e:72a7:f484:432e:7a59:929b  
Link-local IPv6 Address . . . . . : fe80::549a:37c2:ff57:9fb3%3  
IPv4 Address. . . . . : 10.220.185.190  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::7080:bcff:fe6a:e650%3  
                             10.220.185.174
```

Fig 4. Current IP Address, Subnet Mask, Default Gate way of my mobile device

3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access. Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

```
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Pinging 192.168.1.1 with 32 bytes of data:  
  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255  
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255  
  
Ping statistics for 192.168.1.1:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms  
  
C:\>
```

Fig 5. DHCP Configuration result

Interface: FastEthernet0	
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.1.50
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::202:4AFF:FE02:C01B
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5

Fig 6. Static IP Allocation

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
|
```

Fig 7. Static Configuration Result

4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

If two devices are assigned the same IP address, an **IP address conflict** occurs. Both devices may face communication problems and may not be able to access the network properly because the router cannot identify the correct device.

Current Range is: **192.168.1.100 - 192.168.1.149**. – 50 Devices

Physical
Config
Desktop
Programming
Attributes

Web Browser
X
URL http://192.168.1.1/
Go
Stop

Internet Setup
Internet Connection type
Automatic Configuration - DHCP
Optional Settings (required by some internet service providers)
Host Name:
Domain Name:
MTU:
Size: 1500

Network Setup
Router IP
IP Address:
192
.
168
.
1
.
1
Subnet Mask:
255.255.255.0
DHCP Server Settings
DHCP Server:
☒ Enabled
☐ Disabled
DHCP Reservation
Start IP Address: 192.168.1. 100
Maximum number of Users: 50
IP Address Range: 192.168.1. 100 - 149
Client Lease Time: 0 minutes (0 means one day)
Static DNS 1: 0 . 0 . 0 . 0
Static DNS 2: 0 . 0 . 0 . 0
Static DNS 3: 0 . 0 . 0 . 0
WINS: 0 . 0 . 0 . 0

Fig 8. Accessing router admin page